Light

GCSE Computer Science - Network Protocols (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the primary function of a DNS (Domain Name System)?

- **A.** To translate domain names into IP addresses
- **B.** To encrypt all website data
- **C.** To store all website files
- **D.** To create new websites

Q2 (1 mark)

What is a data packet in network communication?

- **A.** A physical envelope containing a hard drive

- ○ **B.** A complete file sent in one piece
- ○ **C.** A type of network cable
- ○ **D.** A small unit of data with header information for routing across a network

Q3 (1 mark)

In packet switching, what happens to packets travelling to the same destination?

- ○ **A.** They must all follow exactly the same route
- ○ **B.** They are always lost during transmission
- ○ **C.** They may take different routes and arrive in different orders
- ○ **D.** They cannot be sent to the same destination

Q4 (1 mark)

TCP is described as a 'connection-oriented' protocol. What does this mean?

- ○ **A.** It requires a physical cable connection
- ○ **B.** It establishes a connection between sender and receiver before transmitting data
- ○ **C.** It can only connect to one device at a time
- ○ **D.** It connects directly to the internet without a router

Q5 (1 mark)

UDP is described as a 'connectionless' protocol. What does this mean?

- ○ **A.** It sends data without establishing a connection first
- ○ **B.** It cannot connect to the internet
- ○ **C.** It only works without cables
- ○ **D.** It deletes all data after sending

Q6 (1 mark)

What is a checksum used for in network protocols?

- ○ **A.** Counting the number of packets sent
- ○ **B.** Encrypting data for security
- ○ **C.** Speeding up data transmission
- ○ **D.** Detecting whether data has been corrupted during transmission

Q7 (1 mark)

In TCP, what is an acknowledgement (ACK)?

- ○ **A.** A request for the receiver's password
- ○ **B.** The first packet in any transmission
- ○ **C.** A message sent by the receiver confirming that a packet arrived successfully

- ○ **D.** An error message indicating transmission failure

Q8 (1 mark)

Why do packets contain sequence numbers?

- ○ **A.** To count how many people have viewed the data
- ○ **B.** To allow packets to be reassembled in the correct order at the destination
- ○ **C.** To encrypt the packet contents
- ○ **D.** To identify which computer sent the packet

Q9 (1 mark)

In TCP, what happens when the sender does not receive an acknowledgement within the timeout period?

- ○ **A.** The sender retransmits the packet
- ○ **B.** The sender permanently gives up on the transmission
- ○ **C.** The receiver automatically receives the data anyway
- ○ **D.** The network speeds up to compensate

Q10 (1 mark)

What information is typically found in a packet header?

- ○ **A.** Only the actual message content
- ○ **B.** The user's personal details
- ○ **C.** Source address, destination address, and control information
- ○ **D.** Advertisements for products

Q11 (1 mark)

When you type a website address into a browser, which server is contacted first?

- ○ **A.** The web server hosting the website
- ○ **B.** An email server
- ○ **C.** A file server
- ○ **D.** A DNS server to find the IP address

Q12 (1 mark)

What is meant by 'protocol overhead'?

- ○ **A.** The physical weight of network equipment
- ○ **B.** The extra data added by protocols for control purposes, reducing space for actual content
- ○ **C.** The cost of buying network software
- ○ **D.** The height of server racks

Q13 (2 marks)

A user types 'www.school.ac.uk' into their browser. Describe what happens during DNS resolution.

- ○ **A.** The browser contacts a DNS server, which looks up the domain name and returns the corresponding IP address so the browser can connect to the web server
- ○ **B.** The browser randomly guesses the IP address
- ○ **C.** The website automatically appears without any lookup process
- ○ **D.** The user must manually enter the IP address

Q14 (2 marks)

Why would an online multiplayer game choose UDP over TCP for transmitting player position updates?

- ○ **A.** UDP is more reliable and ensures every packet arrives
- ○ **B.** UDP has lower latency because it does not wait for acknowledgements; slightly outdated positions are acceptable, but delays are not
- ○ **C.** TCP cannot be used for games under any circumstances
- ○ **D.** UDP provides better encryption for game data

Q15 (2 marks)

A large image file is transmitted across the internet. Packets 1, 2, 3, 4, and 5 are sent but arrive in the order 3, 1, 5, 2, 4.
How does the receiving computer correctly reconstruct the image?

- ○ **A.** It displays the image in the wrong order, causing corruption

○ **B.** It requests the sender to resend all packets in the correct order

○ **C.** It is impossible to reconstruct the image if packets arrive out of order

○ **D.** It uses sequence numbers in packet headers to reorder packets 1, 2, 3, 4, 5 before reassembly

Q16 (2 marks)

A packet arrives with a checksum that does not match the calculated checksum of its contents.
What does this indicate and what happens next?

○ **A.** The packet is from an unauthorised sender and is forwarded to authorities

○ **B.** The checksum calculation was too slow and the packet is processed anyway

○ **C.** The packet data was corrupted during transmission; the packet is discarded and retransmission may be requested

○ **D.** The sender used the wrong protocol and must switch to a different one

Q17 (2 marks)

TCP uses a 'three-way handshake' to establish connections.
What is the purpose of this process?

○ **A.** To synchronise sender and receiver, confirm both are ready, and establish connection parameters before data transfer

○ **B.** To transfer three packets of data quickly

○ **C.** To encrypt data three times for extra security

- ○ **D.** To connect exactly three computers together

Q18 (2 marks)

A user visits the same website multiple times throughout the day. After the first visit, subsequent visits load faster.
How does DNS caching contribute to this improvement?

- ○ **A.** DNS caching has no effect on loading speed
- ○ **B.** The entire website is downloaded and stored during the first visit
- ○ **C.** DNS caching only works for email, not websites
- ○ **D.** The IP address is stored locally after the first lookup, eliminating the need to query DNS servers for subsequent visits

Q19 (2 marks)

Voice over IP (VoIP) phone calls transmit voice data across the internet. Why might VoIP applications prefer UDP over TCP for voice data?

- ○ **A.** UDP provides clearer audio quality through error correction
- ○ **B.** TCP cannot transmit audio data
- ○ **C.** Voice communication needs low latency; waiting for retransmissions would cause unacceptable delays in conversation
- ○ **D.** UDP automatically improves poor internet connections

Q20 (2 marks)

A network administrator notices that sending many very small files across the network is inefficient.

Why does protocol overhead make small transfers less efficient?

- ○ **A.** Small files are automatically deleted by the network
- ○ **B.** Each packet requires headers regardless of data size; for small files, headers may be larger than the actual data
- ○ **C.** Small files cannot use network protocols
- ○ **D.** Protocol overhead only affects large files

Q21 (2 marks)

When connecting to a website using HTTPS, the browser checks the website's digital certificate.
What does this verification confirm?

- ○ **A.** That the website is who it claims to be and the connection can be trusted
- ○ **B.** That the website loads quickly
- ○ **C.** That the website has no viruses
- ○ **D.** That the website is free to access

Q22 (2 marks)

When you send an email to someone using a different email provider (e.g., Gmail to Outlook), how does SMTP handle this?

- ○ **A.** The email cannot be delivered between different providers
- ○ **B.** The email is converted to a different format for each provider
- ○ **C.** Both users must use the same SMTP server

○ **D.** Your mail server uses SMTP to relay the email to the recipient's mail server, which then stores it for retrieval

Q23 (2 marks)

What is the difference between bandwidth and latency in network communications?

○ **A.** They are the same thing with different names

○ **B.** Bandwidth measures security strength; latency measures encryption speed

○ **C.** Bandwidth is the amount of data that can be transmitted per second; latency is the time delay for data to travel from source to destination

○ **D.** Bandwidth only applies to wireless; latency only applies to wired

Q24 (3 marks)

A company's DNS server fails. Employees report they cannot access websites by typing addresses like 'www.google.com', but some employees who recently visited those sites can still access them. Analyse why this is happening and what it reveals about DNS.

○ **A.** The websites have been deleted from the internet

○ **B.** Employees who recently visited sites have cached DNS results locally, while others cannot translate domain names to IP addresses without the DNS server

○ **C.** The internet is completely down for everyone

○ **D.** DNS failure has no impact on web browsing

Q25 (3 marks)

A software developer is creating an application that will be used for both file downloads and live video streaming.
Analyse which transport protocol should be used for each feature and justify the choices.

- ○ **A.** TCP for file downloads (reliability essential - every byte must arrive correctly) and UDP for live streaming (low latency essential - occasional frame loss acceptable)
- ○ **B.** UDP for both because it is always faster
- ○ **C.** TCP for both because reliability is always more important than speed
- ○ **D.** Neither TCP nor UDP - a new protocol must be created

Q26 (3 marks)

A network has multiple possible routes between two cities. One major route fails due to a cable break.
Analyse how packet switching allows communication to continue despite this failure.

- ○ **A.** All communication stops until the cable is repaired
- ○ **B.** Packets are independently routed; routers detect the failure and automatically direct packets via alternative routes, maintaining connectivity
- ○ **C.** Packet switching only works when all routes are functioning
- ○ **D.** Users must manually select alternative routes for their data

Q27 (3 marks)

A company transmits large backup files nightly over their network. They notice TCP overhead consumes significant bandwidth.
Analyse the trade-offs of TCP overhead for this specific use case.

- ○ **A.** TCP overhead has no impact on backup efficiency
- ○ **B.** The company should never use TCP for large files
- ○ **C.** UDP would provide identical reliability with no overhead
- ○ **D.** TCP overhead (headers, acknowledgements, retransmissions) reduces effective throughput, but ensures backup integrity; for critical data like backups, reliability justifies the overhead cost

Q28 (3 marks)

An attacker sets up a fake Wi-Fi hotspot and attempts to intercept users' banking credentials.
Analyse how HTTPS protects users even when connected to a malicious network.

- ○ **A.** HTTPS cannot protect users on malicious networks
- ○ **B.** HTTPS only works on trusted networks
- ○ **C.** HTTPS encrypts all data end-to-end and verifies the bank's identity via certificates; the attacker cannot read encrypted traffic or impersonate the bank without a valid certificate
- ○ **D.** The attacker can easily bypass HTTPS by changing the URL

Q29 (3 marks)

Trace the complete journey of an email from when a user clicks 'Send' until the recipient reads it on their phone.
Identify all protocols involved at each stage.

- ○ **A.** SMTP sends the email from client to sender's mail server, then to recipient's mail server; IMAP or POP3 retrieves it to the recipient's phone; DNS resolves server addresses throughout
- ○ **B.** Only one protocol (HTTP) handles the entire process
- ○ **C.** FTP transfers the email as a file between computers
- ○ **D.** The email travels directly from sender to recipient without any servers

Q30 (3 marks)

New wireless transmission technology is developed that is ten times faster than current Wi-Fi. Existing applications continue to work without modification.
Analyse how the layered protocol model enables this seamless adoption.

- ○ **A.** All applications must be completely rewritten for new transmission technology
- ○ **B.** Layered models prevent adoption of new technology
- ○ **C.** The new technology operates at the Network Access layer only; upper layers (Internet, Transport, Application) remain unchanged, so applications using those layers continue working without modification
- ○ **D.** Transmission technology changes require new protocols at every layer

Total: **55** marks • Time: **30 min**

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